

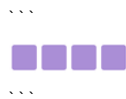


Now, we have four empty squares remaining in a straight line, which matches the shape of Tetromino I. Let's look at the variations of Tetromino I:

Variation 1:



Variation 2:



Since the remaining empty squares are in a horizontal line, Variation 2 of Tetromino I will fit perfectly. Let's place Variation 2 of Tetromino I into the target rectangle:



The correct variation of Tetromino I that fits into the target rectangle is:

B. 2

(b) Inconsistency between language and visualization - Page 2

Figure 16: Inconsistency between language and visualization

Let's start by visualizing the target rectangle with the empty squares (□) and the existing polyominoes:

...



...

We have two polyominoes to fit into this rectangle: Tetromino L (■) and Tetromino I (■). We need to determine which variation of Tetromino L will fit without overlapping any existing polyominoes or going outside the rectangle.

Let's consider the variations for Tetromino L:

Variation 1:

...



...

Variation 6:

...



...

We also have the variations for Tetromino I, but since it's a straight line, it can only fit in one way in the remaining space after placing Tetromino L.

Let's try to fit Variation 1 of Tetromino L first:

...

(a) Example case of Self-refine - Page 1



...

As we can see, Variation 1 of Tetromino L does not fit because it overlaps with the existing yellow (■) polyomino.

Now let's try Variation 6 of Tetromino L:

...



...

Variation 6 of Tetromino L does not fit either because it goes outside the rectangle.

Since neither Variation 1 nor Variation 6 of Tetromino L fits into the target rectangle without overlapping existing polyominoes or going outside the rectangle, the correct answer is:

C. Neither

However, there seems to be a mistake because the task guarantees that there always exists a solution. Let's re-evaluate the placement of Tetromino L.

We can see that the Tetromino L must fit in the top left corner because that's the only place with enough space for its shape. Let's try placing Variation 1 of Tetromino L in the top left corner:

...

(b) Example case of Self-refine - Page 2

Figure 17: Example case of Self-refine